

Dongzhi Luo

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EDUCATION

City University of Hong Kong <i>Incoming M.Sc. in Physics with Data Modelling and Quantum Technologies</i>	Sep 2026 - Jul 2027 (expected)
Lanzhou University <i>Graduate Study in Territorial Spatial Planning</i>	Sep 2024 - Jul 2025 GPA: 87.6/100
Northeastern University <i>Bachelor's Degree in Urban and Rural Planning</i>	Sep 2019 - Jul 2024 GPA: 79.3/100

RESEARCH INTERESTS

Computational urban science; AI-assisted urban planning; agent-based modelling; human mobility; spatial optimization

PUBLICATIONS

PUBLISHED

Qiao, W., & Luo, D.* (2025). An autoencoder-based framework for analyzing regional variations in urban green space demand. *Frontiers in Sustainable Cities*, 7, 1642184. <https://doi.org/10.3389/frsc.2025.1642184>

MANUSCRIPTS

Luo, D., Zhang, D., Yang, G., & Wu, C.* Staged optimization and bounded semantic reasoning for urban planning. Manuscript under review at *Environment and Planning B: Urban Analytics and City Science*.

Wu, C., Luo, D., Mingshu, W.* Geographic coverage masks uneven lead authorship in coastal sustainability research. Manuscript in preparation for *Nature Sustainability*.

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (Guangzhou) <i>Research Assistant Guangzhou, China</i>	Aug 2025 - Aug 2026
- Investigated AI-assisted urban planning, spatial optimization, and agent-based modelling; translated planning questions into reproducible Python and GIS workflows. - Developed spatial-analysis and simulation pipelines for layout generation, mobility analysis, and interactive research prototypes; contributed to manuscripts, figures, and collaborative project delivery.	

SELECTED RESEARCH PROJECTS

Research Geography and Lead Authorship in Coastal Sustainability Project Lead	Dec 2025 - Present
Constructed a reproducible evidence-mapping workflow comparing study locations with lead-author affiliations; developed a geographic-coverage versus research-leadership framework and a six-figure Analysis manuscript for <i>Nature Sustainability</i>.	
Staged Optimization and Bounded Semantic Reasoning for Urban Planning Project Lead	Aug 2025 - Jun 2026
Designed a staged NSGA-II framework that separates land-use structure generation from indicator calibration and uses bounded LLM screening for semantic feasibility; exported feasible Pareto solutions for preference experiments.	
Dual-Environment Cognitive Urban Agent Simulator Project Lead	2026 - Present
Built GIS-derived actual and cognitive environments with heterogeneous routines, partial observation, and dynamic awareness, familiarity, trust, and uncertainty updates; implemented interactive SVG graph and map diagnostics.	
Older-Adult GPS Map Matching and Activity-Route Extraction Research Contributor	2026 - Present
Designed a quality-gated HMM/Viterbi workflow and implemented person-day route extraction for 1,369,928 GPS points from 105 participants; produced validated GIS datasets and an offline interactive web map.	

SELECTED RESEARCH PROJECTS (CONTINUED)

Functional Enclosure and Public-Space Accessibility | Research Collaborator **Jan 2026 - Aug 2026**

- Contributed to a two-city, AOI-level functional-enclosure index integrating urban morphology, street networks, and aggregate mobility data; supported figure audit and cross-scale reporting for 20,494 residential AOIs.

Urban Green Space Demand Evaluation with Autoencoders | Project Lead **Dec 2024 - Apr 2025**

- Integrated Ga2SFCA accessibility, ecological and social indicators, autoencoder feature extraction, and k-means++ clustering; led Python analysis and manuscript development.

EXPERIMENTAL APPS AND GAME PROTOTYPES

Starsea Embers - 2D Pixel Strategy Game | Independent Developer **2026**

- Built a playable Python prototype with a procedurally generated 24-system galaxy, multi-faction AI, resource, colony, fleet, wormhole, and heat-death endgame systems; validated core logic across 30 random galaxies.

TOEFL Vocabulary Trainer | Independent Developer **2026**

- Developed and packaged a standalone Windows learning application with a 777-item question bank, per-user progress persistence, and a reproducible build pipeline.

Origami-Themed Puzzle Game | Collaborative Design Project **2026**

- Contributed to concept development and a warm, stylized visual and promotional direction linking gameplay, art references, and audience communication.

PROFESSIONAL EXPERIENCE

Liaoning Urban and Rural Construction Planning Design Institute | Assistant Planner **Nov 2023 - Mar 2024**

- Supported county-level territorial spatial planning through baseline research, land-use indicators, spatial-data organization, technical reports, and regulatory drawing review.

AWARDS

3rd Prize, China Graduate Smart City Technology & Creative Design Competition **Sep 2025**

Third-Class Academic Scholarship, Lanzhou University **2024, 2025**

3rd Prize, Rural Revitalization Design Competition, Liaoning Civil Architecture Society **Sep 2023**

Outstanding Student Leader, Northeastern University **Dec 2020**

TECHNICAL SKILLS

Programming and Analysis: Python, MATLAB, SPSS; machine learning

Geospatial: ArcGIS, QGIS, GeoPandas

Agent and LLM Systems: agent-based modelling, LLM-simulated agents

Development and Design: Figma, Adobe tools

Languages: Chinese (native); English (CET-6: 462)